



AER1416: Numerical Methods for Uncertainty Quantification

University of Toronto Institute for Aerospace Studies
<http://utias.utoronto.ca/~pbn>

Assignment 2 (40pts)

Due April 9, 2026

1. **[6pts]** Apply Latin Hypercube Sampling (LHS) to approximate the first two statistical moments of the output in problems **P1** ($d = 10$), **P2** and **P3**. Consider only the case when the underlying random variables are uniformly distributed. A MATLAB code for LHS can be downloaded from the folder **Software** on Quercus.
2. **[6pts]** Apply the Quasi Monte Carlo method based on Sobol sequences to approximate the first two statistical moments of the output in problems **P1** ($d = 10$), **P2** and **P3**. Consider only the case when the underlying random variables are uniformly distributed. A MATLAB code for generating Sobol sequences for the unit cube $[0, 1]^d$ can be downloaded from the folder **Software** on Quercus.¹
3. **[12pts]** Apply the sparse quadrature scheme (Clenshaw-Curtis) to approximate the first two statistical moments of the output in problems **P1** ($d = 10$), **P2** and **P3**. Consider only the case when the underlying random variables have a uniform distribution. A MATLAB library for generating the quadrature nodes and weights for the sparse tensor product Clenshaw Curtis rule can be downloaded from the folder **Software** on Quercus.²
4. **[6pts]** Consider the n -dimensional integral $I = \int \phi(\mathbf{x})p(\mathbf{x})d\mathbf{x}$, where $\phi : \mathbb{R}^n \rightarrow \mathbb{R}$ and p denotes the joint probability density function (pdf) of the vector of random variables $\mathbf{x}$. Unfortunately, the joint pdf is only known up to a constant, i.e., $p(\mathbf{x}) = q(\mathbf{x})/Z$, where $q : \mathbb{R} \rightarrow \mathbb{R}$ is a known function and Z is an unknown scalar normalization constant.

Formulate an importance sampling scheme to estimate I . Clearly show all steps in your derivation.

¹The original source of this MATLAB library is http://people.sc.fsu.edu/~jburkardt/m_src/sobol/sobol.html which also contains additional documentation. Python version of this library can be found at https://people.sc.fsu.edu/~jburkardt/py_src/sobol/sobol.html

²The original source of this MATLAB library is http://people.sc.fsu.edu/~jburkardt/m_src/sparse_grid_cc/sparse_grid_cc.html which contains additional documentation.

5. [5pts] Read Mike Giles' seminal paper "Multilevel Monte Carlo Path Simulation"³ Your response should be around 1-2 pages in length (not smaller than 10pt font with 1-inch margins).
 - Explain the key insight behind multilevel Monte Carlo methods
 - For the case when there are two levels, highlight the similarities and differences with the control variate approach we covered in class.
 - Discuss how multilevel Monte Carlo methods could be leveraged to solve the assignment test-cases in AER1416.
6. (5pts) Study the paper by O'Hagan, "Monte Carlo is fundamentally unsound," *J. R. Statist. Soc. D*, 36(2/3), 1987, pp. 247-249 <http://doi.org/10.2307/2348519>. Your response should be around 1-2 pages in length (not smaller than 10pt font with 1-inch margins).
 - (a) Explain O'Hagan's central critique of Monte Carlo methods. What philosophical inconsistency does he identify in how function information is utilized?
 - (b) Among the methods we studied in this course, which of the approaches seem most aligned with addressing O'Hagan's concerns? Explain your reasoning with specific examples.

Guidelines:

- Your report must not exceed 14 pages (with at least 10pt font type and single spacing).
- Suggested structure of report: Introduction and problem statement, methodology, structure of your code, results and discussion, and concluding remarks. Dedicate one section to each problem and present results in tables and/or figures as appropriate.
- While using LHS and Sobol sequences, increase the sample size and estimate the number of samples required to achieve good approximations.
- For the sparse quadrature scheme, study the convergence of results as the quadrature level is increased.
- Compare the results obtained using LHS, Quasi Monte Carlo and sparse quadrature schemes to those obtained earlier in Assignment 1 using the classical Monte Carlo method. Clearly summarize the findings from your numerical comparison studies and draw relevant conclusions.
- Submit an **electronic copy** of your report in PDF format, **documented** MATLAB scripts and **input files** (if any). You should include a file named "README" outlining how the scripts should be run. Upload a single **tar** or **zip** file containing all files to Quercus. You are expected to verify the integrity of your tar/zip file before uploading it to Quercus.

³Download link: https://people.maths.ox.ac.uk/gilesm/files/OPRE_2008.pdf